

Nikolai Ovchinnikov

Senior C++17/20/23 Engineer — Low-Latency / ULL Trading and Backend Systems

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Summary

~10 years building performance-sensitive C++ and backend systems on Linux, from low-level FPGA/digitizer tooling to high-load storage and trading infrastructure. Strongest in modern C++, production debugging, concurrency, and systems that sit close to hardware, networking, or large-scale data paths.

Experience

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|--|---|----------------|
| Software Engineer, Trading Systems | Maven Securities | 2025 - Present |
| <ul style="list-style-type: none">Build modern C++ components for FPGA-accelerated, ultra-low-latency options execution on Linux, close to market data, order routing, risk checks, and production deploymentShipped CBOE C2 (US options) support end-to-end, spanning exchange connectivity, FPGA integration, market-data configuration, packaging, and deploy specsExtended the hardware execution path for complex/combo orders, including message encoding, checksum/feedback handling, and FPGA regression coverageImplemented delta-aware volume capping and order-volume fixes across CBOE/CME execution logic, translating trading-risk constraints into latency-sensitive C++ codeStack: C++17/20/23, Boost (Hana/Asio), Cap'n Proto, component-graph metaprogramming, CMake, Linux; FPGA offload (ExaNIC, Xilinx XDMA) | | |
| Software Engineer, Data Privacy | Booking.com | 2022 - 2025 |
| <ul style="list-style-type: none">Led migration of a critical PII-storage service from bare-metal to Kubernetes — 400K RPS at ~90% lower costDrove DMA compliance: RFC, implementation, rollout monitoring — zero client-facing disruptionStack: 🐍 Java (Spring Boot), Apache Cassandra + Kafka, Kubernetes, Linux; | | |
| Software Engineer, Mail | Yandex | 2021 - 2022 |
| <ul style="list-style-type: none">Built C++17/20 backend services on one of Europe's largest PostgreSQL installations; integrated AWS SDK for C++ with Boost.Asio coroutinesShipped archive/restore of user mail data — ~10% PostgreSQL storage saved | | |
| Software Engineer | Guzik Technical Enterprises | 2019 - 2021 |
| <ul style="list-style-type: none">Built C/C++17 software for high-throughput analog-to-digital converters end-to-end (low-level HW → UI); led the Python/CPython extension for the digitizer SDKRebuilt FPGA CI/CD — 2-3× faster pipelines with better failure diagnostics | | |

Education

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| Novosibirsk State University | Novosibirsk, Russia |
| Bachelor of Science in Computer Science, GPA 4.5 / 5.0 | 2015 - 2019 |
| <ul style="list-style-type: none">Thesis (C++17 / OpenCV): stereo-camera calibration and 3D reconstruction — applied computer vision and numerical optimization | |

Skills

- Programming Languages:** 🇨🇧 C++17/20/23 (primary). 🇨🇦 Go, 🐍 Python, 🐚 Bash — tooling/scripting
- Tools & Systems:** low-latency / ULL systems, FPGA offload (ExaNIC, Xilinx), lock-free & concurrency, async I/O, Boost (Hana/Asio), Cap'n Proto / Omnibus messaging, CMake, Linux, profiling, production debugging; PostgreSQL, Docker, Cython, AI/LLM.
- Collaboration:** onboarding new joiners across teams, mentoring junior/mid engineers, technical writing.

Other

Languages: English (proficient), Russian (native), Dutch (A2)

Side Interests: Reverse engineering & CTF prize-winner; hobby electronics (STM32, ESP32, Arduino); kitesurfing, padel.

Freelance software developer alongside studies — **100+ commercial projects** delivered end-to-end